Baseline: ConvNeXt-Tiny for Rice Leaf Diseas Classification

Config: ImageNet- K pretrained | GELU activation | Stochastic Depth | LayerNorm **Model:** Cor Tiny

Dataset: Rice Leaf Disease Identification Dataset

. Colab Setup (jalankan sekali per session)

In [ ]:
```
from google.colab import drive
drive.mount('/content/drive')
import os
DRIVE_DIR = '/content/drive/MyDrive/rice-convnext/baseline_tiny'
os.makedirs(DRIVE_DIR, exist_ok=True)
print(f'Drive folder ready: {DRIVE_DIR}')
```

```
Mounted at /content/drive
Drive folder ready: /content/drive/MyDrive/rice-convnext/baseline_tiny
```

In [ ]:
```
from google.colab import files
import os, shutil
print('Upload kaggle.json...')
files.upload()
os.makedirs('/root/.kaggle', exist_ok=True)
shutil.move('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 0o600)
print('Kaggle API ready!')
```

```
Upload kaggle.json...
```

Upload widget is only available when the cell has been executed in the current browser session. Pl rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
Kaggle API ready!
```

In [ ]:
```
DATASET_DIR = '/content/drive/MyDrive/rice-convnext/dataset'
if os.path.exists(DATASET_DIR) and len(os.listdir(DATASET_DIR)) > 0:
    print(f'Dataset sudah ada, skip download.')
else:
    !pip install kaggle -q
    !kaggle datasets download -d wangxiaoqii/rice-leaf-disease-identifica
    !unzip -q /content/tmp/*.zip -d {DATASET_DIR}
    !rm -rf /content/tmp
    print('Dataset ready!')

for root, dirs, files_list in os.walk(DATASET_DIR):
    level = root.replace(DATASET_DIR, '').count(os.sep)
    if level > 1: break
    indent = '  ' * level
    print(f'{indent}{os.path.basename(root)}/ ({len(files_list)} files)')
```

```
Dataset sudah ada, skip download.
dataset/ (0 files)
  Bacterial Blight/ (0 files)
```

In [ ]:
```python
import shutil

LOCAL_DATASET = '/content/dataset'

if not os.path.exists(LOCAL_DATASET):
    print('Copying dataset ke local SSD...')
    shutil.copytree(DATASET_DIR, LOCAL_DATASET)
    print('Done!')
else:
    print('Sudah ada di local.')

DATA_DIR = LOCAL_DATASET
```

```
Copying dataset ke local SSD...
Done!
```

. Setup & Imports

In [ ]:
```python
import os, random, shutil
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
from sklearn.metrics import classification_report, confusion_matrix
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms, models

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
torch.backends.cudnn.deterministic = True

DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {DEVICE}')
if torch.cuda.is_available():
    print(f'GPU: {torch.cuda.get_device_name(0)}')
```

```
Device: cuda
GPU: Tesla T4
```

. Configuration

In [ ]:
```python
DATA_DIR     = '/content/drive/MyDrive/rice-convnext/dataset'
DRIVE_DIR    = '/content/drive/MyDrive/rice-convnext/baseline_tiny'
IMG_SIZE     = 224
BATCH_SIZE   = 64
NUM_EPOCHS   = 30
LR_INIT      = 1e-3
```

```
LR_FINAL     = 1e-6
WEIGHT_DECAY = 0.05
NUM_WORKERS  = 2
IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD  = [0.229, 0.224, 0.225]
SAVE_PATH    = f'{DRIVE_DIR}/convnext_convnext_tiny_baseline.pt'
EXP_NAME     = 'ConvNeXt-Tiny | GELU | StochDepth | LayerNorm'
print(f'Experiment: {EXP_NAME}')
```

```
Experiment: ConvNeXt-Tiny | GELU | StochDepth | LayerNorm
```

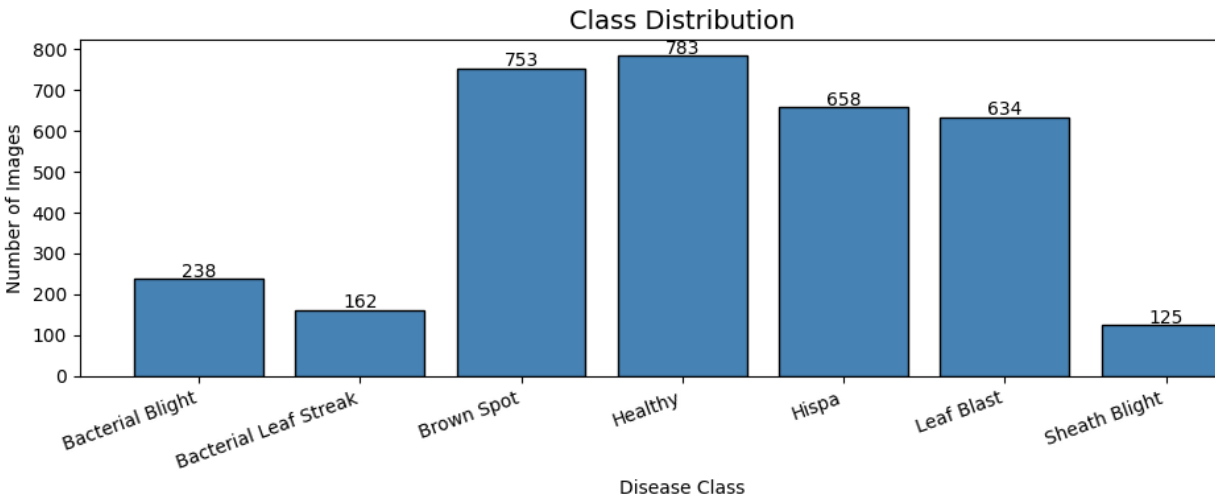
. Data Loading & Augmentation

In [ ]:
```
train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])
val_test_transforms = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(IMG_SIZE),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])

full_dataset = datasets.ImageFolder(DATA_DIR)
CLASS_NAMES  = full_dataset.classes
NUM_CLASSES  = len(CLASS_NAMES)
print(f'Classes ({NUM_CLASSES}): {CLASS_NAMES}')
print(f'Total images: {len(full_dataset)}')

total   = len(full_dataset)
n_train = int(0.6 * total)
n_val   = int(0.2 * total)
n_test  = total - n_train - n_val

generator = torch.Generator().manual_seed(SEED)
train_set, val_set, test_set = random_split(full_dataset, [n_train, n_val

train_set.dataset.transform = train_transforms
val_set.dataset.transform   = val_test_transforms
test_set.dataset.transform  = val_test_transforms

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False
test_loader  = DataLoader(test_set,  batch_size=BATCH_SIZE, shuffle=False
print(f'Train: {len(train_set)} | Val: {len(val_set)} | Test: {len(test_s
```

```
Classes (7): ['Bacterial Blight', 'Bacterial Leaf Streak', 'Brown Spot', 'Hea
'Hispa', 'Leaf Blast', 'Sheath Blight']
Total images: 3353
Train: 2011 | Val: 670 | Test: 672
```

In [ ]:
```
labels_all = [full_dataset.targets[i] for i in range(len(full_dataset))]
counts = [Counter(labels_all)[i] for i in range(NUM_CLASSES)]
```

```
plt.figure(figsize=(10, 4))
bars = plt.bar(CLASS_NAMES, counts, color='steelblue', edgecolor='black')
plt.title('Class Distribution', fontsize=14)
plt.xlabel('Disease Class')
plt.ylabel('Number of Images')
plt.xticks(rotation=20, ha='right')
for bar, count in zip(bars, counts):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+5, str(cou
plt.tight_layout()
plt.savefig('class_distribution.png', dpi=150)
plt.show()
```



. Model — ConvNeXt-Tiny

In [ ]:
```
model = models.convnext_tiny(weights=models.ConvNeXt_Tiny_Weights.IMAGENE
in_features = model.classifier[2].in_features
model.classifier[2] = nn.Linear(in_features, NUM_CLASSES)
model = model.to(DEVICE)
total_p     = sum(p.numel() for p in model.parameters())
trainable_p = sum(p.numel() for p in model.parameters() if p.requires_gra
print(f'Total params     : {total_p:,}')
print(f'Trainable params: {trainable_p:,}')
```

```
Downloading: "https://download.pytorch.org/models/convnext_tiny-983f1562.pth"
root/.cache/torch/hub/checkpoints/convnext_tiny-983f1562.pth
100%|██████████| 109M/109M [00:00<00:00, 179MB/s]
Total params     : 27,825,511
Trainable params: 27,825,511
```

. Training Setup

In [ ]:
```
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=LR_INIT, weight_decay=WEIG

def get_scheduler(optimizer, num_epochs, lr_init, lr_final):
    warmup = optim.lr_scheduler.LinearLR(optimizer, start_factor=0.01, en
    cosine = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=num_ep
    return optim.lr_scheduler.SequentialLR(optimizer, schedulers=[warmup,

scheduler = get_scheduler(optimizer, NUM_EPOCHS, LR_INIT, LR_FINAL)
```

. Training Loop

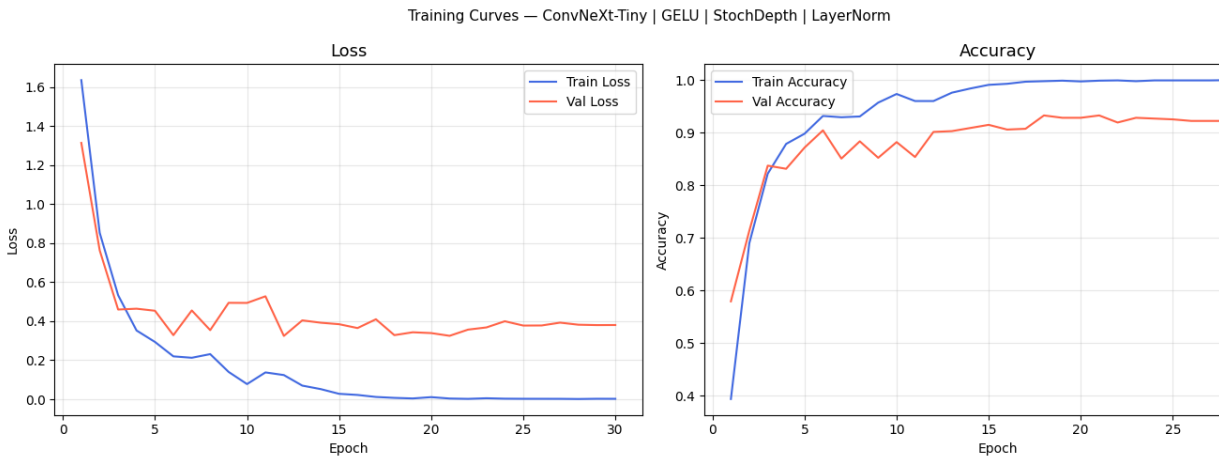
In [ ]:
```
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for images, labels in loader:
        images, labels = images.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * images.size(0)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total   += labels.size(0)
    return running_loss / total, correct / total

def evaluate(model, loader, criterion):
    model.eval()
    running_loss, correct, total = 0.0, 0, 0
    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = outputs.max(1)
            correct += predicted.eq(labels).sum().item()
            total   += labels.size(0)
    return running_loss / total, correct / total

history = {'train_loss': [], 'train_acc': [], 'val_loss': [], 'val_acc':
best_val_acc = 0.0

for epoch in range(NUM_EPOCHS):
    train_loss, train_acc = train_one_epoch(model, train_loader, criterio
    val_loss,   val_acc   = evaluate(model, val_loader, criterion)
    scheduler.step()
    history['train_loss'].append(train_loss)
    history['train_acc'].append(train_acc)
    history['val_loss'].append(val_loss)
    history['val_acc'].append(val_acc)
    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), SAVE_PATH)
    print(f'Epoch [{epoch+1:02d}/{NUM_EPOCHS}] '
          f'Train Loss: {train_loss:.4f} Acc: {train_acc:.4f} | '
          f'Val Loss: {val_loss:.4f} Acc: {val_acc:.4f}'
          + (' <- best' if val_acc == best_val_acc else ''))

print(f'\nBest Val Accuracy: {best_val_acc:.4f}')
print(f'Saved: {SAVE_PATH}')
```

```
Epoch [01/30] Train Loss: 1.6356 Acc: 0.3933 | Val Loss: 1.3140 Acc: 0.5791 <
Epoch [02/30] Train Loss: 0.8527 Acc: 0.6897 | Val Loss: 0.7627 Acc: 0.7134 <
Epoch [03/30] Train Loss: 0.5332 Acc: 0.8220 | Val Loss: 0.4598 Acc: 0.8373 <
Epoch [04/30] Train Loss: 0.3524 Acc: 0.8787 | Val Loss: 0.4641 Acc: 0.8313
Epoch [05/30] Train Loss: 0.2937 Acc: 0.8981 | Val Loss: 0.4536 Acc: 0.8716 <
Epoch [06/30] Train Loss: 0.2198 Acc: 0.9319 | Val Loss: 0.3284 Acc: 0.9045 <
Epoch [07/30] Train Loss: 0.2126 Acc: 0.9294 | Val Loss: 0.4551 Acc: 0.8507
Epoch [08/30] Train Loss: 0.2313 Acc: 0.9309 | Val Loss: 0.3541 Acc: 0.8836
Epoch [09/30] Train Loss: 0.1400 Acc: 0.9572 | Val Loss: 0.4942 Acc: 0.8522
Epoch [10/30] Train Loss: 0.0777 Acc: 0.9736 | Val Loss: 0.4937 Acc: 0.8821
Epoch [11/30] Train Loss: 0.1375 Acc: 0.9602 | Val Loss: 0.5276 Acc: 0.8537
Epoch [12/30] Train Loss: 0.1239 Acc: 0.9602 | Val Loss: 0.3245 Acc: 0.9015
Epoch [13/30] Train Loss: 0.0699 Acc: 0.9761 | Val Loss: 0.4042 Acc: 0.9030
Epoch [14/30] Train Loss: 0.0520 Acc: 0.9841 | Val Loss: 0.3923 Acc: 0.9090 <
Epoch [15/30] Train Loss: 0.0278 Acc: 0.9910 | Val Loss: 0.3846 Acc: 0.9149 <
Epoch [16/30] Train Loss: 0.0221 Acc: 0.9930 | Val Loss: 0.3649 Acc: 0.9060
Epoch [17/30] Train Loss: 0.0117 Acc: 0.9970 | Val Loss: 0.4099 Acc: 0.9075
Epoch [18/30] Train Loss: 0.0071 Acc: 0.9980 | Val Loss: 0.3285 Acc: 0.9328 <
Epoch [19/30] Train Loss: 0.0041 Acc: 0.9990 | Val Loss: 0.3433 Acc: 0.9284
Epoch [20/30] Train Loss: 0.0108 Acc: 0.9975 | Val Loss: 0.3392 Acc: 0.9284
Epoch [21/30] Train Loss: 0.0037 Acc: 0.9990 | Val Loss: 0.3250 Acc: 0.9328 <
Epoch [22/30] Train Loss: 0.0020 Acc: 0.9995 | Val Loss: 0.3568 Acc: 0.9194
Epoch [23/30] Train Loss: 0.0051 Acc: 0.9980 | Val Loss: 0.3681 Acc: 0.9284
Epoch [24/30] Train Loss: 0.0029 Acc: 0.9995 | Val Loss: 0.3997 Acc: 0.9269
Epoch [25/30] Train Loss: 0.0024 Acc: 0.9995 | Val Loss: 0.3777 Acc: 0.9254
Epoch [26/30] Train Loss: 0.0022 Acc: 0.9995 | Val Loss: 0.3781 Acc: 0.9224
Epoch [27/30] Train Loss: 0.0021 Acc: 0.9995 | Val Loss: 0.3926 Acc: 0.9224
Epoch [28/30] Train Loss: 0.0012 Acc: 1.0000 | Val Loss: 0.3820 Acc: 0.9224
Epoch [29/30] Train Loss: 0.0025 Acc: 0.9995 | Val Loss: 0.3800 Acc: 0.9239
Epoch [30/30] Train Loss: 0.0022 Acc: 0.9995 | Val Loss: 0.3804 Acc: 0.9224

Best Val Accuracy: 0.9328
Saved: /content/drive/MyDrive/rice-convnext/baseline_tiny/
convnext_convnext_tiny_baseline.pt
```

. Training Curves

In [ ]:

```
epochs_range = range(1, NUM_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(epochs_range, history['train_loss'], label='Train Loss', col
axes[0].plot(epochs_range, history['val_loss'],   label='Val Loss',   col
axes[0].set_title('Loss', fontsize=13)
axes[0].set_xlabel('Epoch'); axes[0].set_ylabel('Loss')
axes[0].legend(); axes[0].grid(alpha=0.3)
axes[1].plot(epochs_range, history['train_acc'], label='Train Accuracy',
axes[1].plot(epochs_range, history['val_acc'],   label='Val Accuracy',
axes[1].set_title('Accuracy', fontsize=13)
axes[1].set_xlabel('Epoch'); axes[1].set_ylabel('Accuracy')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.suptitle(f'Training Curves — {EXP_NAME}', fontsize=11)
plt.tight_layout()
plt.savefig('training_curves.png', dpi=150)
plt.show()
```



. Evaluation on Test Set

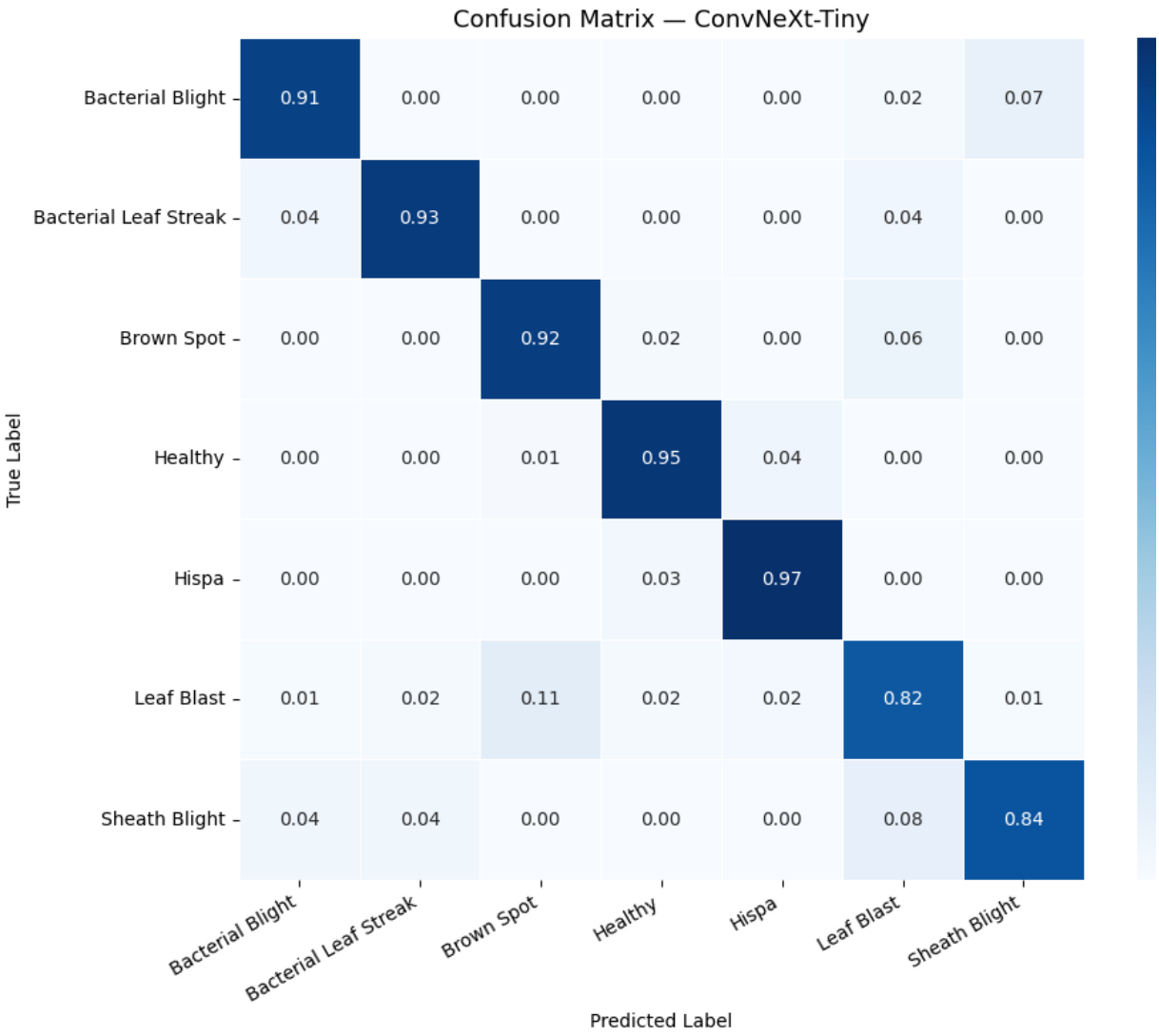
In [ ]:
```
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
all_preds, all_labels = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(DEVICE)
        outputs = model(images)
        _, predicted = outputs.max(1)
        all_preds.extend(predicted.cpu().numpy())
        all_labels.extend(labels.numpy())

report = classification_report(all_labels, all_preds, target_names=CLASS_
report_df = pd.DataFrame(report).transpose()
print('=== Classification Report ===')
print(report_df.round(4).to_string())
report_df.to_csv('classification_report.csv')

summary = {
    'Model'     : ['ConvNeXt-Tiny'],
    'Activation': ['GELU'],
    'Dropout'   : ['Stochastic Depth'],
    'Norm'      : ['LayerNorm'],
    'Accuracy'  : [report['accuracy']],
    'Precision' : [report['weighted avg']['precision']],
    'Recall'    : [report['weighted avg']['recall']],
    'F1-Score'  : [report['weighted avg']['f1-score']],
}
summary_df = pd.DataFrame(summary)
print('\n=== Summary ===')
print(summary_df.round(4).to_string(index=False))
summary_df.to_csv('summary.csv', index=False)
```

```
=== Classification Report ===
                      precision   recall  f1-score   support
Bacterial Blight         0.9423  0.9074    0.9245   54.0000
Bacterial Leaf Streak    0.8966  0.9286    0.9123   28.0000
Brown Spot               0.9107  0.9217    0.9162  166.0000
Healthy                  0.9515  0.9458    0.9486  166.0000
Hispa                    0.9145  0.9727    0.9427  110.0000
Leaf Blast               0.8783  0.8211    0.8487  123.0000
Sheath Blight            0.8077  0.8400    0.8235   25.0000
accuracy                 0.9137  0.9137    0.9137    0.9137
macro avg                0.9002  0.9053    0.9024  672.0000
weighted avg             0.9136  0.9137    0.9133  672.0000

=== Summary ===
        Model Activation          Dropout      Norm  Accuracy  Precision  Rec
F1-Score
ConvNeXt-Tiny       GELU Stochastic Depth LayerNorm    0.9137     0.9136  0.9
0.9133
```

. Confusion Matrix

In [ ]:

```python
cm = confusion_matrix(all_labels, all_preds)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
plt.figure(figsize=(10, 8))
sns.heatmap(cm_norm, annot=True, fmt='.2f',
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES,
            cmap='Blues', linewidths=0.5)
plt.title(f'Confusion Matrix — ConvNeXt-Tiny', fontsize=13)
plt.ylabel('True Label'); plt.xlabel('Predicted Label')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=150)
plt.show()
```



. Save All to Drive

In [ ]:
```
local_files = ['training_curves.png', 'confusion_matrix.png',
               'classification_report.csv', 'summary.csv', 'class_distrib
for fname in local_files:
    if os.path.exists(fname):
        shutil.copy(fname, f'{DRIVE_DIR}/{fname}')

all_outputs = [
    (f'{DRIVE_DIR}/convnext_convnext_tiny_baseline.pt', 'Model weights'),
    (f'{DRIVE_DIR}/training_curves.png',                 'Training curves
    (f'{DRIVE_DIR}/confusion_matrix.png',                'Confusion matri
    (f'{DRIVE_DIR}/classification_report.csv',           'Per-class metri
    (f'{DRIVE_DIR}/summary.csv',                         'Summary table')
]
print(f'=== Output Files: {DRIVE_DIR} ===')
for fpath, desc in all_outputs:
    exists = '✅' if os.path.exists(fpath) else '❌'
    print(f'{exists}  {os.path.basename(fpath):<40} {desc}')
```

```
=== Output Files: /content/drive/MyDrive/rice-convnext/baseline_tiny ===
✅  convnext_convnext_tiny_baseline.pt       Model weights
✅  training_curves.png                      Training curves
✅  confusion_matrix.png                     Confusion matrix
✅  classification_report.csv                Per-class metrics
✅  summary.csv                              Summary table
```